

Measure an actuator on one joint first

M7 • 04 Apr 2027 - 30 Apr 2027 • 75 hours

Main deliverable

Over the month, produce a single-actuator test rig and a model relating torque, speed, current and temperature. The result answers a specific question: can the selected actuator perform the work of a leg or finger joint over the required motion and load range? A catalogue torque at stall does not mean the actuator can hold the robot's body continuously. Define requirements, measure an available sample, separate known parameters from estimates and validate the model on new data. Decide whether to buy a complete actuator set only after testing one sample.

What must already be ready

From M6, bring the controller, synchronised telemetry, error counters, timestamps, command limits, physical stop and verified prevention of automatic restart. From M5, bring link geometry, mountings and approximate mass. Identify where angle is measured: at the rotor, after the gearbox or by a separate output-link sensor. This determines whether gearbox backlash is observable. Entry check: convert degrees to radians, revolutions per second to rad/s and grams to kilograms, and explain force versus torque. Calculate the static torque of a known mass about an axis.

Select equipment around the available actuator and its documentation. You need a securely mounted sample, compatible driver or the smart servo's built-in control interface, angle/current/temperature measurement, a current-limited supply and a guarded fixture. If these are unavailable, first complete requirements and a digital model; clearly separate calculated characteristics from measurements. A bracket may be made on the existing 3D printer after checking geometry, material and load. A printed part does not automatically have the strength of an industrial mounting.

Scope of the month

Study the differences between DC, BLDC and smart servo architectures, but experiment with one type. You do not need to develop your own FOC driver, rewind a motor or buy 12/18 actuators. A smart servo's internal loops may be closed to the user; measure its external behaviour and explicitly identify inaccessible internal signals. Low speed, holding and backlash matter for a hand; loaded torque, repeatability and heating matter for a leg. The method is shared, while requirements differ.

The final report must show the operating region supported by measurements and the modes for which data remains insufficient. This plan specifies no universal current, temperature, voltage or mass limits: derive them from the motor, driver, gearbox, supply and mounting documentation.

How to allocate 75 hours

Balance study and experiments

Hours 1-20: actuator comparison, requirements and preliminary model; 10 hours theory and 10 hours practice. Hours 21-40: measurement fixture, encoder, current, temperature and rig checks; 10+10. Hours 41-60: control loops, limits and repeatable response; 10+10. Hours 61-75: identification, validation on new data and model boundaries; 5+10. Total: 35 hours of reading, calculation and analysis, and 40 hours of practice. Count evaluation of raw measurements as practice, and equation derivation and experiment selection as analysis.

Divide each ten-hour theory block into five 2-hour sessions. In each session, study one mechanism, calculate its effect for your joint, then prepare an experiment. Schedule ten-hour practical blocks on Sundays, divided into stages with breaks; 10 hours means active work. The final 5 theory hours are split 2+2+1. Dates on the first page are the periods in the original overall plan. Shared boundary dates, rest, holidays and integration reviews are unchanged; place steps in available study hours without double counting.

A separate record for each experiment

State motor model and revision, gearbox, firmware, controller gains, supply voltage, ambient temperature, mounting, current sensor and temperature measurement location. Record command and measured angle, speed, current, voltage and temperature on one time axis, or document synchronisation. The commands file contains what was actually sent after limiting, rather than only the original desired profile. Save stop causes and missing-data intervals so fault periods remain visible in plots.

Repository organisation

Create requirements/ for joint calculations, fixture/ for CAD and assembly, acquisition/ for collection code, experiments/ for configurations, data/raw/ for raw runs, models/ for parameters and reports/ for plots and limitations. Split data into training (train) and validation sets by whole runs before fitting. Randomly splitting one run into nearly identical neighbouring training and validation points is not independent validation. Begin with a simple model and retain it as the baseline before adding nonlinearities.

If time is short, reduce the number of brands compared, omit an additional actuator type and simplify the interface. Requirements, measurement path, motion limits, thermal observation and validation on an independent run remain compulsory. Keep missing hardware measurements as open items rather than replacing them with arbitrary numbers.

Joint requirements and architecture selection

What to study

Compare brushed DC, BLDC with a separate driver, and smart servos with integrated control. Consider motor, gearbox, sensor and interface separately: a convenient bus does not guarantee suitable torque, and a powerful motor does not solve feedback problems. Study the torque-speed curve, efficiency, rotor inertia, permissible gearbox load, encoder resolution, backlash and transmission compliance. Distinguish continuous and short-duration operation, and electrical, thermal and mechanical limits.

Task 1. Load profiles for a leg and finger

Start with the M5 mass and geometry; express unknowns as ranges. For a link with a horizontal lever arm, use $\tau = mgr$, with m in kg, g in m/s^2 , r the perpendicular lever arm in metres and τ in $N \cdot m$. 1. Draw a free body with force direction. 2. Calculate three link positions and two payload cases. 3. For the leg, add a support scenario with load distribution explicitly assumed; do not divide robot mass by leg count without describing the support phase. 4. For the finger, use desired fingertip force and lever arm. Save the requirements table and sensitivity to mass and length. Check that doubling the lever arm at constant force doubles static torque. Edge case: zero force lever arm.

Task 2. Compare actuators under relevant operating conditions

Select three architectures or available candidates with primary documentation. Build a comparison table: torque in the required mode, loaded speed, mass, reduction, sensor, backlash, current control, diagnostics, supply, interface and unknown characteristics. 1. Reject any option that fails a hard requirement. 2. Separately assess uncertainty from missing data. 3. Choose one sample for the rig and describe an experiment that could disprove that choice. Prepare a decision justifying each criterion; an arbitrary weighted score is no substitute. Check that every important conclusion has a source, calculation or future test, and that stall torque has not been used as continuous torque.

Reduction and inertia

Fix the definition $N = \text{motor speed/output-shaft speed}$. In an ideal transmission, output speed is divided by N , torque is multiplied by N , and load inertia reflected to the motor is divided by N^2 . Actual efficiency and backlash impose additional limitations. For the chosen geometry, calculate how increasing the ratio changes speed and reflected inertia. These calculations are hypotheses for measurement; gear tooth counts alone cannot establish actual holding accuracy.

Measurement rig and calibration

Task 3. Fixture for one actuator

The aim is repeatable motion without movement of the motor housing. Inputs: mounting drawing, load direction and the sample's permissible operating characteristics. 1. Draw the load path from shaft through mounting to base. 2. Position travel stops, motion-area guarding, access to the stop control and cable strain relief. 3. Check alignment and free travel without powered motion. 4. Perform the first powered run at the minimum justified command, increasing it in predefined steps. Deliver CAD, photographs, a fastener list and the verification record. Edge case: the cable must not become taut and alter measured torque at the travel limits.

Create one table of permissible values for the supply, driver, wires, connectors and fuse. The weakest component determines the limit, considering operating mode, cooling and documentation. A fuse and a software current limit serve different purposes. Separately establish behaviour during braking and energy return to the supply: a supply capable of delivering energy may be unable to absorb it during braking. Do not command sharp acceleration or braking until this behaviour is understood.

Task 4. Encoder angle and speed

Check counts per revolution, quadrature decoding mode, gear ratio and sign. 1. Apply a known manual displacement with the actuator disabled, if its construction permits back-driving. 2. Compare counts with output-link angle. 3. Estimate velocity from angle differences over a known interval and, alternatively, from time between edges if pulses are available. 4. Compare noise and delay at low and higher speeds. Deliver the conversion factor and plots of both estimates. Check that a full revolution or available reference angle produces the expected count. Edge cases: overflow, reversal, zero speed and a missed edge.

Task 5. Current and temperature

Establish what the sensor measures: supply current, winding current or current in the control coordinate frame. These quantities are not automatically interchangeable as I in $\tau = K_t I$. Measure zero-current offset, compare known points with a reference instrument and retain uncertainty. Record temperature with the sensor location and settling time: housing and windings heat differently. Deliver calibration coefficients and a description of limitations. Fault: a missing or saturated instrument must not permit continuation of an experiment that depends on its readings.

Position control and bounded commands

Control-loop architecture

Determine who closes the current, velocity and position loops: your code, the driver or smart-servo firmware. Use SimpleFOC and ODrive resources to draw the cascade and label the input and output units of each block. An external controller must not duplicate an unknown internal loop without understanding its mode. PID relates error to control action; feedforward predicts the required action. Command limiting introduces nonlinearity: the integrator can continue accumulating error when the actuator can no longer meet the request.

Task 6. Repeatable position response

Initial conditions: a checked rig, known limits and a small motion profile away from stops. 1. Begin with an available simple mode and a reproducible initial state. 2. Apply a smoothly changing command with specified limits; save position, speed and actual control action. 3. Change one gain or profile parameter while keeping load constant. 4. Repeat the best configuration several times. Deliver reference/measurement plots, steady-state error, overshoot and settling time. Define the settling band beforehand in angle units; measure the time until the response stays within it. Edge case: reversal and return to zero.

Task 7. Saturation and anti-windup

Begin in simulation of the simple joint. 1. Limit available control action. 2. Apply a command that cannot be achieved quickly. 3. Compare an ordinary integrator with the chosen anti-windup method: integrator limiting or back-calculation. 4. Restore an achievable reference and measure recovery time. Save two plots of control action and integrator state under identical conditions. Check that you can explain why controller gains cannot remove a physical limit. Confirm on hardware only with a smooth motion profile inside the operating region; deliberately jamming the shaft to demonstrate saturation is unnecessary.

Task 8. Motion profile and stopping

Set position, speed and acceleration limits from rig requirements and documentation. Generate a bounded trajectory between two angles and compare it with an abrupt step in simulation. Use only the permissible profile on the rig. Measure maximum speed, current, error and behaviour near the software limit. Reject or limit an out-of-range command under an explicit rule and log it; a mechanical stop is not a routine way to end every motion. Communication loss and restart use the tested M6 state machine, while the mechanism's subsequent state depends on its load and construction.

Identification: friction, delay and backlash

A testable model

Start at the output joint: $J\ddot{q} + b\dot{q} + \tau_c \cdot \text{sign}(\dot{q}) + \tau_g(q) = \tau_a$. Here J is inertia about the axis in $\text{kg}\cdot\text{m}^2$, b is viscous resistance in $\text{N}\cdot\text{m}\cdot\text{s}/\text{rad}$, τ_c approximates dry friction, τ_g is gravitational torque and τ_a is actuator torque. Near zero speed, the sign model is inadequate: rest, breakaway and backlash require separate observations. If torque is unavailable, use the stock actuator command as input and identify the closed-loop command-to-angle system, explicitly stating that the model includes the actuator and its internal controller.

Task 9. Backlash and dead zone

The aim is to understand why a finger ignores a small command and why a leg reverses with a delay. 1. Reproduce slow motion in both directions over one range. 2. Record input command and angles before and after the transmission if both sensors are available. 3. Plot output against input in both directions and identify the region where input changes produce no measurable output. 4. Repeat with another permissible load. Save dead-zone and backlash estimates with sensor resolution. If only the motor encoder is available, output-link mechanical backlash cannot be claimed as directly measured; a separate observation method is required.

Task 10. Steps and frequency excitation

Prepare small bounded steps and a smooth chirp within the region authorised beforehand; chirp means excitation frequency changes over time. First calculate maximum profile speed and acceleration to ensure they remain within limits. 1. Record raw responses. 2. Identify delay and characteristic response time. 3. Fit a minimal model to the first group of runs. 4. Validate it with another profile and initial position. Save parameters, applicability limits, raw data and residuals. Do not increase frequency into dangerous oscillation merely to extend the measured bandwidth.

Why not fit everything at once?

Friction, delay, an inaccurate torque constant and a velocity filter can explain similar errors. Check units and measurement delay first, then a simple linear model; add one mechanism only when its effect has independent evidence. Save parameters with units, estimation method and a confidence range or observed spread. Lower error on the training record is not success if a new record is predicted worse. Show one case the model reproduces and one where it predictably loses accuracy.

Torque, speed, heating and the operating region

Task 11. A verifiable torque characteristic

If an independent force sensor and known lever arm are available, estimate torque as $\tau = F \cdot r_{\text{perp}}$. Begin with a secured rig, small justified loads and known geometric accuracy. 1. Check sensor zero. 2. Measure several safe operating conditions with different loads and speeds. 3. Repeat in both directions. 4. Compare torque with measured current, accounting for current type and transmission. Deliver characteristic points with variability and operating limits. If torque is not independently measured, give a calculated estimate and a method for later verification; do not call a smart servo's internal "load" indicator an accurate torque sensor without manufacturer confirmation.

Task 12. Thermal experiment with controlled duration

Choose a repeated profile resembling leg work or finger holding, within documented limits. 1. Begin from a known thermal state. 2. Record ambient and available-sensor temperatures, current, operating mode and fraction of time under load. 3. Stop at the preset duration or an earlier documentation-based stop condition. 4. Record cooling after the experiment. Save temperature rise above ambient and load duty cycle. A short record showing continued heating does not establish an acceptable steady-state thermal regime.

The simplest thermal model relates temperature rate of change to dissipated power and heat removal. In the first version, estimating characteristic heating and cooling times and explaining applicability limits is sufficient. The I^2R loss approximation requires the appropriate current and resistance; supply current and unknown internal commutation may not provide this information. Do not infer unmeasured winding temperature from an external sensor without a justified thermal-coupling model.

A combined operating-range map

Plot measured points in the speed-torque plane, marking current, temperature, duration and limiting factor. Display catalogue limits separately from your measurements. For a leg, state the supported profile and tested duration; for a finger, separately assess holding and repeated small motions. Check repeatability after cooling. An actuator with a different gearbox, voltage or cooling arrangement is a different configuration requiring verification. The final map informs robot design; it does not guarantee every future operating mode.

Fault tests and numerical criteria

Measurement path

F1: encoder sign reversed. Detect the error in a software test and an initial small motion before continuing closed-loop control; record expected and actual direction. F2: incorrect counts per revolution. Comparison with an independent angle must reveal the scale error. F3: current or temperature data disappears. Mark the run invalid and stop it if the relevant parameter is a safety condition for the experiment. Do not reuse the last available value as fresh protective information. F4: timestamps jump or frames are lost. Do not treat angle differentiation across a corrupted interval as reliable velocity.

Control and mechanics

F5: target angle beyond the software limit. Check the chosen command rejection or limiting rule; log requested and applied commands. F6: control action saturates in simulation. After an achievable reference returns, anti-windup restores normal operation according to a defined metric; retain the saturation interval in the report. F7: communication loss during permissible motion. Test the M6 stopping procedure and prevention of automatic restart. F8: small reversal. Compare the expected dead zone with instrument resolution; the program must not infer absence of mechanical motion from a single unchanged count.

Validate the model on another record

F9: a model fitted to one profile receives another profile, load or amplitude within its declared region. Compare angle RMSE, maximum absolute error and transition-time error. $RMSE = \sqrt{\text{mean}((q_{\text{model}} - q_{\text{measured}})^2)}$; its unit matches the angle unit. F10: load or temperature goes beyond the identification region in simulation or existing data. The report must explicitly label extrapolation rather than conceal it in an overall mean error. F11: power restoration. Motion must not resume solely from a saved reference.

Set thresholds before the experiment

Choose angle tolerance from the joint's future function and measurement capability; do not specify "0.1 degree" without reference to the encoder and backlash. Define a settling band and required time within it before measuring settling time. Compare model error with a simple baseline, such as a first-order system with constant delay; a more complex model must improve independent validation. For each experiment, retain maximum measured current and temperature with duration, sample rate and measurement location. Preserve a failed run with its stop cause and do not assess it as a fully completed run.

First 40 hours: requirements and rig

Hours 1-10: determine what must be measured

1-2: break leg and finger requirements into position, speed, torque and operating duration. 3-4: study DC, BLDC and smart servo architectures; draw a motor-gearbox-sensor-controller diagram. 5-6: calculate task 1 static torques and their geometric dependence. 7-8: study the torque-speed characteristic, reflected inertia and efficiency; plot simple expected trends. 9-10: read documentation for available candidates and list unknown parameters. Finish by selecting the question the rig experiment must answer.

Hours 11-20: document requirements and prepare the experiment

11-12: complete task 1 load scenarios and unit table. 13-14: build the task 2 comparison table with sources and characteristic conditions. 15-16: create a simple joint simulation with bounded control action; save parameters and hypotheses. 17-18: design the task 3 fixture geometry and check load transfer, stop access and cables. 19-20: prepare test commands, acquisition configuration and the first experiment procedure. If the sample is not yet known, remain at interface level and explicitly mark parameters TBD; do not copy another motor's limits.

Hours 21-30: measurement and protection

21-22: study the supply, driver, wiring, fuse and braking behaviour from the exact components' documentation. 23-24: study the encoder and velocity-estimation methods; calculate the smallest observable angle change. 25-26: establish current measurement location and type, sensor offset and range. 27-28: study thermal modes, temperature-sensor location and housing/winding temperature differences. 29-30: prepare calibration procedures, initial states and stop conditions. Finish this block with a usable measurement procedure, rather than notes alone.

Hours 31-40: first physical dataset

31-32: assemble and check the fixture without powered motion, then perform the first smooth-motion run. 33-34: verify task 4 angle scale and sign. 35-36: compare velocity estimates and retain stationary noise measurements. 37-38: calibrate current and temperature channels for task 5. 39-40: repeat a short profile, check synchronisation between fields, stopping and recreation of the plot from raw data. Saving results is included in every step. Without a real sample, complete equivalent software tests and leave physical acceptance open until the next available session.

Hours 41-75: control and modelling

Hours 41-50: theory through your joint

41-42: draw the actual inner and outer loops; label inputs and units. 43-44: study P, I, D and velocity measurement; explain the effect of each gain. 45-46: study action limits, integral accumulation and anti-windup; prepare a simulation comparison. 47-48: calculate a profile with speed and acceleration limits; distinguish a software boundary from a physical stop. 49-50: define response metrics, repeatability, settling band and communication-fault behaviour. Do not attempt complex tuning until feedback sign and angle scale are verified.

Hours 51-60: control practical

51-52: run the smooth task 6 profile; save the baseline response and several repeats. 53-54: change one control parameter and compare under identical conditions. 55-56: complete task 7 simulation of saturation and recovery. 57-58: verify the task 8 bounded profile on the rig, including a request outside the software range. 59-60: perform task 9 slow reversals; assess dead zone and whether backlash can be measured. Freeze the working configuration at the end; identification runs must not mix different controller gains without labelling them.

Hours 61-65: prepare identification

61-62: derive the simple joint model and choose a measurable input: torque, current with a justified conversion constant or an external smart-servo command. 63-64: divide future runs into fitting and validation sets; choose task 10 excitation and task 11 torque-measurement conditions; calculate permissible profile limits. 65: finalise the task 12 thermal procedure, report format and sample-suitability criterion. Split these 5 hours 2+2+1; all following 10 hours are practice, keeping the total at 75.

Hours 66-75: experiments and independent validation

66-67: record bounded steps and an available frequency profile; estimate delay and initial parameters. 68-69: obtain available torque measurements or an explicitly labelled calculated estimate; verify directions and repeatability. 70-71: run the thermal profile for the predefined duration and also record cooling. 72-73: fit the model and validate on another run; compare residuals with the baseline and explain one systematic error. 74-75: assemble the operating region, model limitations and decision on the suitability of one sample. If time is insufficient to reach steady temperature, state that in the report; a short experiment's results cannot be transferred to continuous operation.

Presenting the result and moving to mechanics

What the final package must contain

Requirements for the chosen joint with load scenarios; architecture comparison and sample justification; CAD and photographs of the guarded rig; supply, stop and limit descriptions with sources; angle, current and temperature calibration; raw synchronised recordings; controller parameters; a torque-speed-current-heating model within available measurements; independent validation and a fault list. For every model parameter, state units, estimation method, operating range and confidence level. Unmeasured torque or winding temperature remains unknown even if software can plot it.

Completion criteria from the original plan: selection rests on measurements, electrical and thermal parameters stay within justified limits, and a set of 12/18 actuators is not selected from one catalogue number. A negative decision is valid: the sample may be unsuitable because of heating, backlash, speed, mass or diagnostics. This is a useful completed result if evidence supports the reasons and requirements for the next sample are stated. Publishing the report here means preparing a reproducible package; no external sharing or posting is required.

Self-check questions

1. Why is supply current not always the current that determines torque? 2. What changes when the reduction ratio doubles? 3. How does backlash differ from elastic deformation? 4. Where is the encoder, and which error can it not observe? 5. Why does a good unloaded response not resolve leg-support requirements? 6. What does the integrator do during saturation? 7. Why can the same PID produce a different response after heating? 8. What exactly constitutes independent model validation? 9. How can measurement delay be distinguished from actuator delay? 10. Why does housing temperature not guarantee a known winding temperature? Each answer needs your own plot, calculation or explicit data limitation.

Handover to M8 and M9

Pass axis geometry, permissible angle ranges, sign, zero, resolution and measured dead zone to M8. Kinematics must generate achievable references within these limits, but does not itself establish sufficient torque. Pass the identified joint, masses, inertia estimates, friction, delay, input interface and independent recordings to M9. Keep motor and link parameters separate to avoid counting the gearbox or inertia twice. When mechanics, load or cooling changes, flag the need for new validation because the previous operating region may change.

Assigned reading: a focused reading route

Books and notes may be in English; keep the plan, definitions and your explanations in English for this edition. Read only the specified sections within the existing 75 hours.

Core reading and one reference route

- 1. Core:** K. Lynch, F. Park, Modern Robotics, §8.9. Actuation, Gearing, and Friction: motor, gearbox and reflected inertia. Read for tasks 1-2 and 9 and explain why motor and output-joint parameters differ.
- 2. Core:** K. J. Åström, R. Murray, Feedback Systems, 2nd edition. Chapter 11 PID Control: Basic Control Functions and Integrator Windup. Read for tasks 6-8; draw your loops and compare saturation with anti-windup. Other editions have different numbering.
- 3. Core, selectively:** R. Tedrake, Underactuated Robotics: System Identification. Sections on least squares and dynamics identification. For tasks 9-10, list measurable inputs, unknown parameters and a separate validation recording; skip complex models.

Documentation for the practical work

ESP-IDF: Pulse Counter - Channel Actions and Glitch Filter for the encoder; SimpleFOC: Current Sensing - motor versus supply current for task 5.

ODrive: Controller - cascade diagram and Trajectory Control for tasks 6-8. This is an architectural example; obtain your driver's limits only from its own datasheet.

Complete the reading with an action

After each section, perform one calculation or short experiment. Save the actuator, driver and instrument datasheet revisions. For tasks 10-12, use the familiar SciPy `least_squares`: check parameter physical meaning and residuals on new data. Buying printed books is optional.

Glossary: terms and definitions

Torque. The turning effect of force: $\tau = r \times F$; torque about a specified axis is the projection of τ onto that axis, in N·m.

Gear ratio. The ratio of input to output speed; here $N = \omega_{\text{motor}} / \omega_{\text{joint}}$, with rotation direction specified separately.

Efficiency. The ratio of useful output power to input power under stated conditions; it depends on speed and load.

Torque constant. Coefficient K_t relating motor electromagnetic torque to current under the adopted model, in N·m/A.

Back EMF. Voltage generated by motor rotation that opposes the supply; proportional to speed in a simple model.

Stall current. Current in a stationary motor at a specified voltage; this is a limiting condition, not an allowable continuous current.

Continuous torque. Torque permissible continuously under specified cooling, speed and temperature conditions; the manufacturer's conditions must be confirmed.

Reflected inertia. Inertia referred through a gearbox to a chosen shaft; rotor inertia at the output scales as N^2 .

Backlash. The relative angular travel between mating components over which a reversal is not immediately transmitted to the load.

Static friction. Contact resistance before motion starts; it takes the required value only up to the maximum holding limit.

Viscous friction. An approximation of motion resistance as torque proportional to angular velocity and opposing it.

Quadrature encoder. A sensor producing two phase-shifted A/B signals from which angle increments and rotation direction can be determined.

Counts per revolution. The number of counted signal transitions per revolution of the specified shaft; it depends on decoding mode and gearbox.

Feedback loop. A path using measured output to change a command in order to reduce a specified control error.

Saturation. Limiting actual control action at a physical or software boundary, even when the calculated command is larger.

Anti-windup. A rule correcting integral state during output limiting, reducing excessive error accumulation and recovery delay.

System identification. Determining model parameters or structure from recorded inputs and outputs under explicit assumptions.

Residual. The difference between measured output and model prediction for the same inputs, time and initial state.

Thermal time constant. A characteristic temperature-change time in the adopted thermal model; it depends on heat capacity and heat transfer.

Check the relationships: $P_{\text{mech}} = \tau \omega$; for an ideal gearbox, $\omega_{\text{out}} = \omega_{\text{in}} / N$ and torque increases by a factor of N . Losses and heating require separate treatment.

Equipment and software for M7

USE EXISTING EQUIPMENT

MacBook Pro M4; tools and the M4 CC/CV supply; one ESP32-S3 from the two M6 boards, IMU, logic analyser and USB cables. The 0-30 V, 5-10 A rating specifies the supply's capabilities: set experiment voltage and current according to the chosen load's limits.

BUY BEFORE RIG WORK: hours 21-40

1 motor: a 12 V brushed DC gearmotor with an A/B quadrature encoder and documented gear ratio and counts. **1 driver:** a bidirectional H-bridge with current limiting, protection and 3.3 V-compatible inputs; select current rating and cooling for the motor and chosen experiments.

1 current-measurement path in the driver, or a separate sensor or shunt with amplifier; **1 contact temperature sensor** with logged readings. **1 fixture set:** rigid mounting, hub or lever, stops, guarding, power switch, fuse and correctly sized wires. Do not pass power-stage current through a solderless breadboard.

Example datasheet study: [Pololu 37D with encoder](#). Check encoder-output voltage; add level conversion before ESP32-S3 if needed. [TI DRV8876](#) illustrates protection and current-sensing requirements; this does not promise compatibility with that motor.

ADDITIONAL ITEMS, IF JUSTIFIED BY CALCULATION

For independent torque measurement in task 11: 1 calibrated force gauge with a suitable range plus a measured lever arm. A separate smart actuator for the future leg: only 1 candidate after calculating requirements. The rig's DC motor does not automatically become the final leg actuator; do not yet order a leg set or 18 actuators.

SOFTWARE AND READINESS CHECK

No new mandatory packages: macOS ARM, pinned ESP-IDF and Python, NumPy, SciPy, Matplotlib and pytest from M6. Build firmware, record one complete revolution in both directions, compare current with a reference, and check temperature, limiting and stopping before loading. Test USB with the actual connection; a VM is not assumed verified.

Preparation is included in the existing 75 hours; arrange equipment delivery in advance and exclude delivery time from study hours.

Motion fundamentals: what a gearbox provides

Required: connect joint requirements with the motor

Angle specifies the desired pose, but holding and acceleration require torque. For a leg, foot load and link weights matter; for a finger, the object's force on a phalanx matters. Before calculating, draw the axis and lever arm: force in N and torque in N·m differ. If the force's line of action passes through the axis, its torque about that axis is zero.

A minimal transmission model

For one revolute joint, q is output angle and $v=\dot{q}$ is velocity in rad/s. Choose shaft signs so that $N=\omega_m/v>0$. Then $v=\omega_m/N$. With power transmitted from motor to load and gearbox efficiency η , output torque is approximately $\tau_{out}=\eta N\tau_m$. Hence $\tau_{out}\cdot v=\eta\tau_m\cdot\omega_m$: a gearbox changes torque and speed but does not create power.

At an illustrative operating point, let $N=20$, $\eta=0.75$, $K_t=0.04$ N·m/A, winding current $i=1$ A and $\omega_m=100$ rad/s. Neglecting losses within the motor itself, $\tau_m=K_t\cdot i=0.04$ N·m, $v=5$ rad/s and $\tau_{out}=0.6$ N·m. Mechanical power is 4 W at the motor shaft and 3 W after the gearbox. Supply consumption requires separate measurement of the electrical path.

Separately draw a horizontal link: x right, y up, axis $+z$ and positive torque anticlockwise. A mass $m=0.5$ kg at lever arm $r=0.1$ m produces gravitational torque $-mgr=-0.4905$ N·m at $g=9.81$ m/s². Rest requires compensation of $+0.4905$ N·m, excluding link weight. The difference $0.6-0.4905=0.1095$ N·m does not by itself establish actuator suitability: operating speed, heating and continuous holding differ.

How inertia affects rapid motion

Acceleration also requires torque $I_{joint}\cdot\ddot{q}$, with I_{joint} in kg·m². Rotor inertia reflected to the output is N^2I_{motor} ; load inertia reflected to the motor is divided by N^2 . High reduction therefore helps produce torque but changes dynamics and sensitivity to contact. Constant η is an approximation for the specified motoring condition; do not transfer it to rest or reverse energy flow without verification.

Checks within existing tasks 1 and 11

Build a calculation table and power balance. In the same exercise model, change N to 40: velocity becomes 2.5 rad/s, torque 1.2 N·m, output power remains 3 W and reflected rotor inertia increases fourfold. Then insert your confirmed M7 parameters and label unknowns separately. Check gravitational torque using $r\times F$ and the existing task 11 experiment; without an independent instrument, retain the result as a calculation.

Included in the 75 hours: 1 hour from 7-8 for transmission and inertia; 1 hour from 11-12 for the load table; 1 hour from 68-69 for checking the existing run. This refines deliverables already prepared. Exercise values are not specifications for the purchased motor and must not be sent as commands.

Reference: Lynch/Park, Modern Robotics, §8.9.1 DC Motors and Gearing; §8.9.2 Apparent Inertia. Read only the relationships between torque, speed, power and reflected inertia; FOC development is not added here.